

Varun Budati

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EDUCATION

Virginia Tech , Blacksburg, Virginia	Aug 2023 – May 2027
B.S. in Computer Science	GPA: 3.59/4.0
Minor in Mathematics & Finance	In-Major GPA: 3.76/4.0
CFA Level 1 Candidate	

COURSEWORK

CS 3114: Data Structures and Algorithms, CS 2505/6: Computer Organization I/II, CS 2114: Software Design & Data Structures, CS 3214: Computer Systems

CORE SKILLS

Programming Languages: Python (5 years), MySQL (2 years), Java, C, HTML/CSS.
Frameworks & Libraries: NumPy, Pandas, Matplotlib, Plotly, Sklearn, Seaborn, SciPy, React, Node.js, Flask.
Developer Tools & OS: Git, Docker, AWS, Linux/Unix.

WORK EXPERIENCE

Summer Research Intern , REACH Lab - Virginia Tech, Blacksburg, Virginia	May 2025 – July 2025
- Conducted a 10-week mixed-methods research study analyzing over 24,500 words from introductory Computer Science course materials at MIT, UC Berkeley, and CalTech to identify pedagogical gaps. - Developed a qualitative & quantitative coding framework and utilized computational text analysis tools (LIWC-22) to thematically categorize hundreds of programming examples and assess their linguistic context. - Synthesized findings to reveal a high prevalence of abstract mathematical problems, providing data-driven recommendations to enhance curricula with more inclusive, real-world applications	
Quantitative Researcher , Dataism Lab for Quantitative Finance - Virginia Tech, Blacksburg, Virginia	October 2024 – Present
- Researching optimal order execution by analyzing the market microstructure of Bitcoin trade data to develop and implement advanced trading strategies. - Constructed benchmark execution algorithms in Python, including VWAP and TWAP, to analyze the market impact and transaction costs of trading Bitcoin. - Engineered a reinforcement learning and neural network architecture (PPO, DDQN) to create an adaptive agent that optimizes trade execution strategies in real-time. - Modeling quantitative performance using statistical methods and Python (NumPy, Pandas, SciPy) to analyze trade execution efficiency and market dynamics.	

PROJECT WORK

Poker Game	January 2024 - March 2024
- Engineered a card handling system using JavaScript arrays and objects to manage a 52-card deck with 4 suits and 13 values. - Programmed 8 distinct game variants through conditional logic and dynamic payable adjustments. - Architected poker hand evaluation algorithms with $O(n \log n)$ time complexity for pattern recognition (pairs, straights, flushes).	
Sports Betting Algorithm & Analytics System	August 2024 – May 2025
- Engineered a quantitative sports prediction model using Python (NumPy, Pandas) to identify statistical edges, achieving an 8,400% return on investment (scaled from \$10 to \$850) over a 6-month period. - Built a real-time data pipeline with sports APIs (Requests) to ingest and process player statistics (Pandas), enabling probability calculations with SciPy and statsmodels. - Developed a performance dashboard (Matplotlib, Seaborn) for visual ROI analysis and integrated an automated risk management system to optimize bankroll allocation using the Kelly Criterion.	

LEADERSHIP/ACTIVITES

Treasurer, FinTech Club , Virginia Tech, Blacksburg, Virginia	October 2024 - Present
- Manage the club's budget and all financial operations to fund initiatives and workshops for a community of 100+ members. - Organized and host speaker events with industry professionals from leading finance and technology firms to create career development and networking opportunities. - Led a faculty-advised research project replicating the foundational Evans & Archer (1968) paper on portfolio diversification. - Modeled risk versus portfolio size using Pandas, NumPy, and SciPy, and presented findings that confirmed unsystematic risk is substantially mitigated with 10-20 assets.	